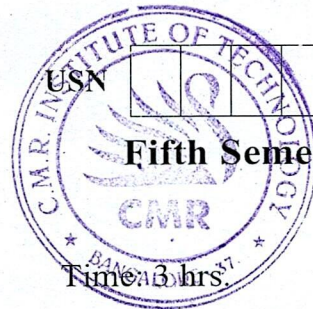


CBCS SCHEME

BAI515B/BCY515B



Fifth Semester B.E./B.Tech. Degree Examination, June/July 2025 Information Retrieval

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.

Module – 1				M	L	C
Q.1	a.	Explain the software architecture of Information Retrieval system.		12	L2	CO1
	b.	Appraise the history of Information Retrieval.		8	L4	CO1
OR						
Q.2	a.	Summarize the features of IR and the problems involved in IR system.		10	L1	CO1
	b.	Explain various methods used in search interfaces for visualization concept.		10	L2	CO1
Module – 2						
Q.3	a.	Illustrate Boolean Model with an example.		10	L3	CO2
	b.	Write a note on weighing and construct term frequency for the document. D1 : To do is to be. To be is to do. D2 : To be or not to be. I am what I am. D3 : I think therefore I am. Do be do be do. D4 : Do do do, da da da, let it be, let it be.		10	L3	CO2
OR						
Q.4	a.	Explain probabilistic IR model in detail.		10	L2	CO2
	b.	Write a note on: i) Latent Semantic Indexing Model ii) Neural Network Model.		10	L1	CO2
Module – 3						
Q.5	a.	What is relevance feedback? List and explain its types.		10	L1	CO3
	b.	Write a note on precision and recall and solve the following. If 20 found documents, 18 relevant, 3 relevant documents are not found, 27 irrelevant are as well not found. Calculate the precision, recall and fallout score for the search.		10	L3	CO3
OR						
Q.6	a.	Describe document pre-processing and the stages in detail.		10	L1	CO3
	b.	Identify different document formats and illustrate their characteristics.		10	L3	CO3

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Module – 4

Q.7	a.	Discuss the structure of inverted indices.	8	L2	CO4
	b.	With a note on suffix tree and construct the suffix tree and suffix tree for the text "Missing Mississippi".	12	L3	CO4

OR

Q.8	a.	Write a note on signature files.	10	L1	CO4
	b.	Explain approximate searching and use dynamic algorithm to search for "colour" in the text "kolorama" and find the minimum errors, time and space complexity.	10	L2	CO4

Module – 5

Q.9	a.	Show the challenges posed by web with respect to search.	10	L4	CO5
	b.	List the components of cluster based architecture and provide a brief description of each.	10	L1	CO5

OR

Q.10	a.	List and explain components of a search engine result page.	10	L1	CO5
	b.	Explain the frame work of a XML retrieval system and its challenges.	10	L2	CO5
